at

meet at equal speeds leftward: do not collide again. At 's second meeting, , give the four displacements from their own equilibria.

Both springs are natural. The four table velocities are, respectively,

right, still: compression speeds up , slows .
rests, moves: stretch speeds up , slows .
The spring oscillators resume oscillating.

From , when

meet again: center-frame velocities by are

at

at

left, right, equally fast: meet without impact. The four table velocities are

Then mass center moves right uniformly at .

move harmonically about at angular frequency .

have mass center . It moves uniformly right at .

do so about at angular frequency . meet every time units, always in either case above: meeting without colliding.